

30/7/26

CO2 - Activity - 2

Q1. CCTV image contains salt-and-pepper

- * Noise: Salt-and-Pepper noise
- * Technique: Median filter
- * It removes black and white impulse noise effectively
- * It preserve image edges better than average filtering

Q2. Satellite image is corrupted by Gaussian noise

- * Noise: Gaussian noise
- * Technique: Gaussian filter
- * It smooths random noise produced by sensors
- * It improves image quality with slight blurring

Q3. Medical X-ray has random noise but edges must be preserved

- * Noise: Random noise
- * Technique: Bilateral filter
- * It removes noise while preserving fine edge
- * It is suitable for medical image enhancement

Q4. Scanned dots:

- * Noise:
- * Technique:
- * It re dots:
- * It re

Q5. Low

- * Noise:
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Q6. Fo

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Q7.

Scanned document has black & white dots:

Q4. Noise: Salt-and-pepper noise

- * Technique: Median filter
- * It removes scattered black and white dots.
- * It keeps text sharp and readable

Q5. Low-light image shows heavy grain

- * Noise: Grain (shot/poisson noise)
- * Technique: Non-Local mean filter
- * It reduces grain without losing details
- * It produces a cleaner low-light image.

Q6. Foggy road image has very low contrast.

- * Problem:- Low contrast
- * Technique:- Histogram equalization
- * It increases overall image contrast
- * It improves visibility in foggy conditions.

Q7. Underwater image appear dull.

- * Problem:- Low contrast and color-loss
- * Technique:- CLAHE
- * It enhances local contrast effectively
- * It restores clearer underwater details.

Q8. Chest X-ray has poor visibility

* Problem: Low contrast

* Technique: CLAHE

* It enhance internal structure clearly

* It avoids excessive brightness changes

Q9. Night-time traffic image lacks intensity variation

* Problem: Low intensity variation

* Technique: Contrast stretching

* It expand the intensity range

* It makes dark object more visible

Q10. Grayscale image uses a narrow intensity range

* Problem: Narrow intensity range

* Technique: Histogram Equalization

* It spreads pixel values across the full range.

* It improves image contrast and clarity.